RAJALAKSHMI ENGINEERING COLLEGE

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V2

Started on	Friday, 7 August 2026, 8:52 AM
State	Finished
Completed on	Friday, 7 August 2026, 9:02 AM
Time taken	9 mins 59 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int c=0;
void factor(int num){
    {
        int i;
        for(i=1;i<=num;++i){
            c++;
            if(num%i==0){
                c++;
            }
            c++;
        }
        c++;
    }
}
int main()
{
    int num;
    scanf("%d",&num);
    factor(num);
    printf("%d",c);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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